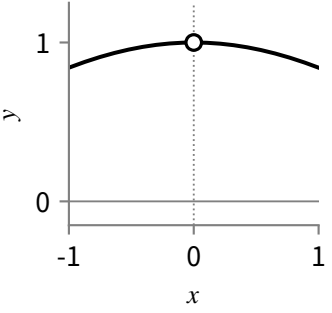
① 可去

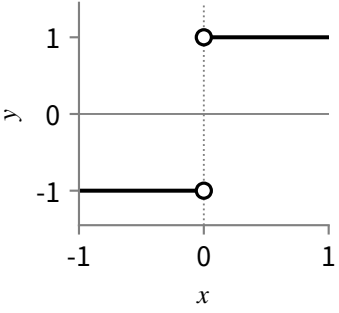
$y = \sin x / x$



缺一点，两侧都趋于 1

② 跳跃

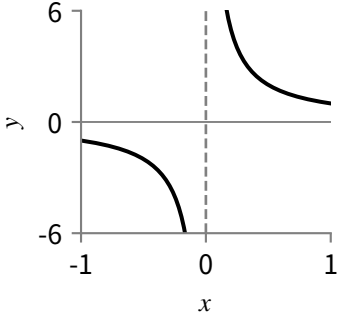
$y = |x| / x$



左 −1，右 1

③ 无穷

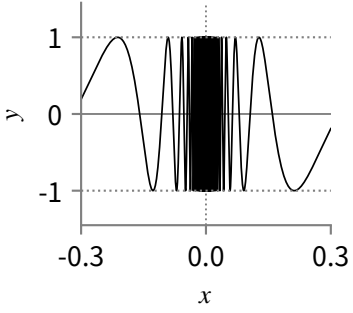
$y = 1/x$



$x = 0$ 是渐近线

④ 振荡

$y = \sin(1/x)$



在 ±1 间来回摆动